

Slot-Attention Guided Hybrid-Driven Extended Object Tracking

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Abstract—Existing Extended Object Tracking (EOT) methods, such as random-matrix and hybrid-driven approaches, often suffer from bias under non-uniform measurements due to single-cluster statistical aggregation. To address this, we propose the slot-attention guided hybrid-driven EOT (SA-HD-EOT). This framework utilizes a Slot-Attention Gaussian Mixture Network (SAGNet) to decompose scans into component-wise soft assignments, which subsequently drive a bank of sub-filters for robust kinematic and extent estimation. Experiments demonstrate that the proposed SA-HD-EOT significantly outperforms the RM-EOT, HD-EOT, and Transformer baselines in both accuracy and stability, while achieving the lowest runtime with moderate memory usage.

Index Terms—Extended object tracking, random matrix, hybrid-driven filtering, slot attention, Gaussian mixture model, non-uniform measurements.

I. INTRODUCTION

EXTENDED object tracking (EOT) aims to jointly estimate an object’s kinematic state and spatial extent from measurements distributed over its surface [1], [2], [3]. Unlike standard point tracking, this paradigm relies on high-resolution sensors capable of yielding multiple detections per scan. Such scenarios are widely encountered in automotive radar/LiDAR perception, marine X-band radar surveillance, and remote sensing [1], [2], [3].

EOT methods broadly fall into random-matrix (RM) recursive filters and set-based deep regressors. A widely adopted formulation models the extent as an ellipse and performs RM filtering with an Inverse-Wishart prior [4], [5], [6], [7], [8]. RM methods offer an attractive trade-off between efficiency, interpretability, and recursive Bayesian estimation, but it is sensitive to partially unknown state-space parameters and distributional assumptions [6], [7], [8], [9]. In contrast, Transformer-style attention mechanisms summarize unordered measurement sets to capture richer spatial patterns [10], [11], [12], yet purely data-driven set-to-state regression often sacrifices interpretability and is challenging for joint regression of dynamics and extent [9].

To address the limitations of purely physical or purely data-driven approaches, model-based deep learning (MBDL) has emerged as a powerful framework that combines the strengths of both [13]. For instance, DNN-aided Kalman

filtering (KalmanNet and its variants) learns latent statistics or correction terms while preserving the recursive structure of classical filters [14], [15], [16]. In the context of EOT, the Hybrid-Driven EOT (HD-EOT) [9] utilizes a learned module (ExONet) to predict uncertain terms governing the gain and extent update, improving robustness to parameter mismatch while retaining the recursive nature of RM filters.

Despite these advancements, physical sensor artifacts, such as non-uniform measurements distribution, challenge the fundamental single-cluster hypothesis of RM-based frameworks [7], [8]. Both RM filters and HD-EOT aggregate the entire scan into a single centroid and scatter matrix [4], [5], [6]. Consequently, these methods suffer from centroid bias towards dense reflection points and extent inflation, failing to decouple local variations from global kinematics. While similar challenges have been addressed in other fields [17], [18], no suitable hybrid-driven solution has been proposed to tackle this issue within the EOT domain.

To address these challenges, we propose the slot-attention guided hybrid-driven extended object tracking (SA-HD-EOT), adopting a decompose–track–fuse framework to mitigate single-cluster bias. We introduce the slot-attention Gaussian mixture network (SAGNet), which parses non-uniform scans into homogeneous soft components, thereby supplying structured observations to a bank of the sub-filters [9]. This design effectively captures multimodal spatial distributions while preserving the computational efficiency of recursive Bayesian estimation. Simulation experiments comparing the SA-HD-EOT against the RM-EOT, Transformer-based methods, and HD-EOT demonstrate its superior robustness, lowest runtime, and moderate memory usage, particularly under challenging, non-uniform measurement scenarios.

II. PROBLEM FORMULATION

We consider track a single extended object in $d = 2$ -dimensional and adopt the random-matrix (RM) formulation, such that kinematics and extent can be estimated recursively with closed-form updates [4], [5], [6]. At each time step k , the kinematic state is $\mathbf{x}_k \triangleq [\mathbf{p}_k^T, \mathbf{v}_k^T]^T$, where $\mathbf{p}_k$ and $\mathbf{v}_k$ denote position and velocity, and the extent is a symmetric positive-definite matrix $\mathbf{X}_k \in \mathbb{R}^{2 \times 2}$. Following [4], [5], [6], the extent is parameterized as $\mathbf{X}_k \triangleq \boldsymbol{\Omega}(\theta_k) \text{diag}(l_{1,k}^2, l_{2,k}^2) \boldsymbol{\Omega}(\theta_k)^T$,

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where $l_{1,k}$ and $l_{2,k}$ are semi-axis lengths and θ_k is the orientation, with $\Omega(\theta_k)$ the rotation matrix [6], [9].

The joint evolution of kinematics and extent follows standard RM dynamics, where the kinematic process noise is coupled with the extent matrix. Specifically, the kinematic state evolves according to a linear Gaussian model [4], [6]

$$\mathbf{x}_k = (\mathbf{F}_k \otimes \mathbf{I}_2) \mathbf{x}_{k-1} + \mathbf{w}_{k-1}, \quad \mathbf{w}_{k-1} \sim \mathcal{N}(\mathbf{0}, \mathbf{D}_k \otimes \mathbf{X}_{k-1}), \quad (1)$$

where $\mathbf{F}_k$ denotes the kinematic transition matrix, $\mathbf{D}_k$ is the process noise covariance, and $\otimes$ represents the Kronecker product [9]. Simultaneously, the extent evolves via a Wishart transition density [6]:

$$p(\mathbf{X}_k | \mathbf{X}_{k-1}) = \mathcal{W}(\mathbf{X}_k; \delta_k, \mathbf{A}_k \mathbf{X}_{k-1} \mathbf{A}_k^T), \quad (2)$$

governed by the degrees of freedom δ_k and the invertible extent evolution matrix $\mathbf{A}_k$ [4], [5], [6].

The standard RM measurement model rely on single-cluster statistical aggregation, where the filter update is derived from the unordered scan set solely through scan-level statistics. At time k , the sensor is assumed to return a raw measurement set $\mathcal{Z}_k \triangleq \{\mathbf{z}_k^j \in \mathbb{R}^2\}_{j=1}^{n_k}$. In this model, each detection $\mathbf{z}_k^j$ is assumed to be

$$\mathbf{z}_k^j = (\mathbf{H}_k \otimes \mathbf{I}_2) \mathbf{x}_k + \mathbf{r}_k^j, \quad \mathbf{r}_k^j \sim \mathcal{N}(\mathbf{0}, \mathbf{B}_k \mathbf{X}_k \mathbf{B}_k^T). \quad (3)$$

Here, $\mathbf{H}_k$ denotes the 1D measurement matrix, and the noise term is governed by $\mathbf{B}_k = (\lambda \mathbf{X}_k + \mathbf{R}_k)^{1/2} \mathbf{X}_k^{-1/2}$ [6], which incorporates the scaling factor λ and the measurement noise covariance $\mathbf{R}_k$. Accordingly, the RM inference depends on $\mathcal{Z}_k$ only through the scan-level sufficient statistics [4], [5], [6]

$$\bar{\mathbf{z}}_k \triangleq \frac{1}{n_k} \sum_{j=1}^{n_k} \mathbf{z}_k^j, \quad \bar{\mathbf{Z}}_k \triangleq \sum_{j=1}^{n_k} (\mathbf{z}_k^j - \bar{\mathbf{z}}_k)(\mathbf{z}_k^j - \bar{\mathbf{z}}_k)^T, \quad (4)$$

implying that RM-EOT and the hybrid-driven variants operate on the tuple $(\bar{\mathbf{z}}_k, \bar{\mathbf{Z}}_k, n_k)$. This strategy is statistically sound only when detections form a near-uniform single cluster. In practical sensing, however, detections are often spatially non-uniform, i.e., concentrated in heterogeneous local regions due to occlusion, glints, multipath, and reflectivity variation [7], [8]. Consequently, such scan-level compression often biases the effective centroid and spread [7], [8].

To capture such non-uniformity, we employ a conditional Gaussian mixture model, where each component corresponds to a local measurement-dense region with specific shifting and scaling relative to the global extent [8]. Formally, for any detection $\mathbf{z}_k^j \in \mathcal{Z}_k$, the likelihood is given by

$$p(\mathbf{z}_k^j | \mathbf{x}_k, \mathbf{X}_k) = \sum_{i=1}^M \pi_{k,i} \mathcal{N}(\mathbf{z}_k^j; \boldsymbol{\mu}_{k,i}, \boldsymbol{\Sigma}_{k,i}). \quad (5)$$

Here, M denotes the number of latent components, and $\{\pi_{k,i}\}_{i=1}^M$ are the mixing coefficients satisfying $\pi_{k,i} \geq 0$ and $\sum_i \pi_{k,i} = 1$. The component-specific mean and covariance are parameterized as

$$\boldsymbol{\mu}_{k,i} = \mathbf{H}_k \mathbf{x}_k + \mathbf{u}_{k,i}, \quad \boldsymbol{\Sigma}_{k,i} = \frac{1}{\tau_{k,i}} \mathbf{X}_k + \mathbf{R}_k, \quad (6)$$

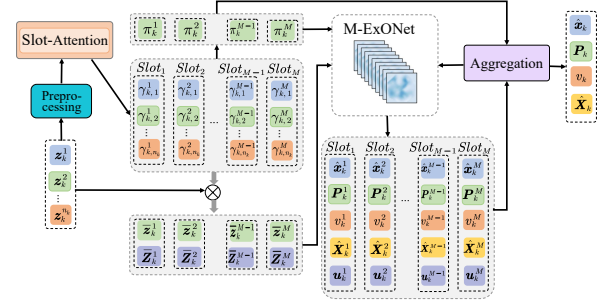


Fig. 1. Pipeline of the SA-HD-EOT

where $\mathbf{u}_{k,i}$ represents the local offset, and the scalar $\tau_{k,i} > 0$ controls the local spread relative to the extent $\mathbf{X}_k$.

The standard scan-level statistics $(\bar{\mathbf{z}}_k, \bar{\mathbf{Z}}_k, n_k)$ fail to characterize the mixture model in (5), as they erroneously conflate distinct latent components. Specifically, the global centroid $\bar{\mathbf{z}}_k$ tends to gravitate toward measurement-dense regions, resulting in biased innovation. Simultaneously, the global scatter $\bar{\mathbf{Z}}_k$ is inflated by an additional between-component term induced by separated component means, which severely skews the extent estimation. Therefore, the core challenge lies in mitigating these biases without sacrificing the computational efficiency and interpretability of the RM framework. This necessity motivates our proposed strategy: a decompose–track–fuse framework that processes distinct latent components via a parallel filter bank to ensure robust tracking.

III. SLOT-ATTENTION GUIDED HYBRID-DRIVEN EOT

The proposed SA-HD-EOT framework integrates a permutation-invariant neural decomposition front-end with a recursive Bayesian filter bank to robustly handle measurement non-uniformity. Instead of compressing a multimodal scan into potentially biased global sufficient statistics, our approach utilizes SAGNet to parse raw, variable-cardinality measurement sets into homogeneous soft components. These component-wise statistics serve as structured inputs for M parallel HD-EOT recursions, whose outputs are subsequently fused to capture complex spatial distributions while maintaining the computational efficiency of the RM-style architecture, as illustrated in Fig. 1.

We employ a permutation-invariant slot attention mechanism to parse the raw scans into explicit soft assignments. Given the measurement set $\mathcal{Z}_k = \{\mathbf{z}_k^j\}_{j=1}^{n_k}$ at time k , SAGNet outputs the soft responsibilities $\{\gamma_{k,i}^j\}_{j=1, i=1}^{n_k, M}$ —representing the probabilistic assignment of the j -th detection to the i -th component—and the mixture weights $\{\pi_{k,i}\}_{i=1}^M$. Specifically, each measurement $\mathbf{z}_k^j \in \mathbb{R}^2$ is mapped to a feature by a shared MLP [19], termed $\mathbf{f}_k^j = \phi(\mathbf{z}_k^j) \in \mathbb{R}^D$, and M learnable slot prototypes are used as queries to match against measurement features $\mathbf{f}_k^j$ through attention [20]. Let $\mathbf{s}_i \in \mathbb{R}^{D_s}$ be the i th slot feature, and define $\mathbf{q}_{k,i} = \mathbf{W}_Q \mathbf{s}_i$, $\mathbf{k}_k^j = \mathbf{W}_K \mathbf{f}_k^j$, $\mathbf{v}_k^j = \mathbf{W}_V \mathbf{f}_k^j$, where $\mathbf{W}_Q, \mathbf{W}_K, \mathbf{W}_V$ are learnable matrix param-

ters. The slot-measurement score is $e_{k,i}^j = \mathbf{q}_{k,i}^T \mathbf{k}_k^j / \sqrt{D_s}$, and responsibilities are obtained by normalizing across slots

$$\gamma_{k,i}^j = \frac{\exp(e_{k,i}^j)}{\sum_{i=1}^M \exp(e_{k,i}^j)}, \quad \sum_{i=1}^M \gamma_{k,i}^j = 1. \quad (7)$$

While $\gamma_{k,i}^j$ routes each measurement to slots, SAGNet also constructs a stable per-slot representation by normalizing across measurements and aggregating values

$$\alpha_{k,i}^j = \frac{\exp(e_{k,i}^j)}{\sum_{r=1}^{n_k} \exp(e_{k,i}^r)}, \quad \mathbf{h}_{k,i} = \sum_{j=1}^{n_k} \alpha_{k,i}^j \mathbf{v}_k^j. \quad (8)$$

A lightweight head then maps $\mathbf{h}_{k,i}$ to mixture parameters $(\pi_{k,i}, \boldsymbol{\mu}_{k,i}, \boldsymbol{\Sigma}_{k,i})$ for likelihood-based pretraining [21], [22].

We bridge the gap between neural soft clustering and recursive Bayesian filtering by converting the raw scan into component-wise statistics via the obtained responsibilities $\{\gamma_{k,i}^j\}_{j=1}^{n_k, M}$. Based on these soft assignments, we compute the following statistics for each component i

$$n_{k,i} \triangleq \sum_{j=1}^{n_k} \gamma_{k,i}^j, \quad \bar{\mathbf{z}}_{k,i} \triangleq \frac{1}{n_{k,i}} \sum_{j=1}^{n_k} \gamma_{k,i}^j \mathbf{z}_k^j, \quad (9)$$

$$\bar{\mathbf{Z}}_{k,i} \triangleq \sum_{j=1}^{n_k} \gamma_{k,i}^j (\mathbf{z}_k^j - \bar{\mathbf{z}}_{k,i})(\mathbf{z}_k^j - \bar{\mathbf{z}}_{k,i})^T, \quad (10)$$

$$\pi_{k,i} \triangleq \frac{n_{k,i}}{\sum_{i=1}^M n_{k,i}}. \quad (11)$$

These statistics effectively isolate distinct measurement-dense regions, thereby removing the between-component scatter inflation and ensuring that each sub-filter receives a statistically sound, unimodal input.

With the multimodal scan decomposed into unimodal statistics, we process each component using a dedicated HD-EOT sub-filter, ensuring that the recursive updates are driven by coherent local geometries rather than a distorted global average. For the i -th component, let $(\hat{\mathbf{x}}_{k|k-1,i}, \hat{v}_{k|k-1,i}, \hat{\mathbf{V}}_{k|k-1,i})$ denote the predicted kinematic and extent parameters derived from (1)–(2). We employ the ExONet module [9] to generate the requisite update parameters. While the standard ExONet outputs the structured factors $(L_{p,k,i}, L_{s,k,i}, \mathbf{B}_{k,i})$ for the Kalman gain and extent update, in the SA-HD-EOT, we modify the network head to additionally regress the component offset vector $\mathbf{u}_{k,i}$.

To integrate the extracted local geometry into the track estimates, we execute the measurement update using an offset-corrected innovation derived from the component-wise statistics. We define the component-wise innovation as

$$\mathbf{g}_{k,i} \triangleq \bar{\mathbf{z}}_{k,i} - (\mathbf{H}_k \otimes \mathbf{I}_d) \hat{\mathbf{x}}_{k|k-1,i} - \mathbf{u}_{k,i}. \quad (12)$$

With $\mathbf{P}_{k|k-1,i} = L_{p,k,i} L_{p,k,i}^T$ and $S_{k,i} = L_{s,k,i} L_{s,k,i}^T$, the kinematic state update is

$$\mathbf{K} \mathbf{G}_{k,i} = \mathbf{P}_{k|k-1,i} \mathbf{H}_{k,i}^T S_{k,i}^{-1}, \quad (13)$$

$$\hat{\mathbf{x}}_{k,i} = \hat{\mathbf{x}}_{k|k-1,i} + (\mathbf{K} \mathbf{G}_{k,i} \otimes \mathbf{I}_d) \mathbf{g}_{k,i}. \quad (14)$$

For the extent, letting $\hat{v}_{k,i} = \hat{v}_{k|k-1,i} + n_{k,i}$, we update the scale matrix as

$$\hat{\mathbf{V}}_{k,i} = \hat{\mathbf{V}}_{k|k-1,i} + \mathbf{N}_{k,i} + \mathbf{B}_{k,i}^{-1} \bar{\mathbf{Z}}_{k,i} \mathbf{B}_{k,i}^{-T}, \quad (15)$$

where $\mathbf{N}_{k,i}$ is the rank-one correction term computed from $\mathbf{g}_{k,i}$ and $L_{s,k,i}$ [9]. The final extent estimate is given by $\hat{\mathbf{X}}_{k,i} = \hat{\mathbf{V}}_{k,i} / (\hat{v}_{k,i} - 2d - 2)$.

To consolidate the multi-component posteriors into a single kinematic and extent estimate, we minimize the Kullback-Leibler (KL) divergence via a moment-matching fusion strategy. Formally, the filter bank induces a mixture posterior $p(\boldsymbol{\xi}_k | \mathcal{Z}_{1:k}) = \sum_{i=1}^M \pi_{k,i} p_i(\boldsymbol{\xi}_k | \mathcal{Z}_{1:k})$ over the latent state $\boldsymbol{\xi}_k$. To retain the tractability of the RM recursion, we approximate the filter-bank posteriors $\{p_i(\boldsymbol{\xi}_k | \mathcal{Z}_{1:k})\}_{i=1}^M$ by a single density $q(\boldsymbol{\xi}_k) \in \mathcal{Q}$ via Kullback-Leibler averaging (KLA): $q^* = \arg \min_{q \in \mathcal{Q}} \sum_{i=1}^M \pi_{k,i} \text{KL}(p_i(\boldsymbol{\xi}_k | \mathcal{Z}_{1:k}) \| q(\boldsymbol{\xi}_k))$ [8].

For the kinematic state, the analytical solution yields the fused mean and covariance

$$\hat{\mathbf{x}}_k = \sum_{i=1}^M \pi_{k,i} \hat{\mathbf{x}}_{k,i}, \quad (16)$$

$$\hat{\mathbf{P}}_k = \sum_{i=1}^M \pi_{k,i} \left(\hat{\mathbf{P}}_{k,i} + (\hat{\mathbf{x}}_{k,i} - \hat{\mathbf{x}}_k)(\hat{\mathbf{x}}_{k,i} - \hat{\mathbf{x}}_k)^T \right), \quad (17)$$

where $\hat{\mathbf{P}}_{k,i} = (\mathbf{I} - \mathbf{K} \mathbf{G}_{k,i} \mathbf{H}_k) \mathbf{P}_{k|k-1,i}$ denotes the standard updated covariance for component i . This fusion explicitly captures the spread of component means, preventing over-confidence. For the extent parameters, we similarly adopt a mixture-weighted fusion

$$\hat{v}_k = \sum_{i=1}^M \pi_{k,i} \hat{v}_{k,i}, \quad \hat{\mathbf{V}}_k = \sum_{i=1}^M \pi_{k,i} \hat{\mathbf{V}}_{k,i}, \quad (18)$$

$$\hat{\mathbf{X}}_k = \hat{\mathbf{V}}_k / (\hat{v}_k - 2d - 2). \quad (19)$$

Additionally, we implement a staged training protocol to establish a stable geometric decomposition before the tracking dynamics are optimized. This reduces optimization coupling: the SAGNet first learns to explain scan geometry via a mixture objective, and the ExONet is then trained on top of a fixed decomposition to avoid degenerate responsibilities early in training. Let T denote the sequence length and $\mathcal{B}$ the mini-batch size in each optimization step.

In the first stage, the SAGNet is pretrained by minimizing the Gaussian-mixture negative log-likelihood [21], [22]:

$$\mathcal{L}_{\text{NLL}} \triangleq - \sum_{k=1}^T \sum_{j=1}^{n_k} \log \left(\sum_{i=1}^M \pi_{k,i} \mathcal{N}(\mathbf{z}_k^j; \boldsymbol{\mu}_{k,i}, \boldsymbol{\Sigma}_{k,i}) \right). \quad (20)$$

Subsequently, we freeze SAGNet and train the ExONet module with the HD-EOT recursion using a supervised tracking loss based on the squared Gaussian Wasserstein distance, which jointly measures kinematic and extent errors

$$\mathcal{L}_{\text{GWD}} \triangleq \frac{1}{\mathcal{B}} \sum_{b=1}^{\mathcal{B}} \sum_{k=1}^T \left(\left\| \mathbf{x}_k^{(b)} - \hat{\mathbf{x}}_k^{(b)} \right\|^2 + \text{tr} \left\{ \mathbf{X}_k^{(b)} + \hat{\mathbf{X}}_k^{(b)} - 2((\mathbf{X}_k^{(b)})^{1/2} \hat{\mathbf{X}}_k^{(b)} (\mathbf{X}_k^{(b)})^{1/2})^{1/2} \right\} \right). \quad (21)$$

In the final stage, optimal end-to-end performance is pursued by unfreezing SAGNet and fine-tuning the full model jointly. We minimize $\mathcal{L}_{\text{GWD}} + \lambda \mathcal{L}_{\text{NLL}}$, where $\mathcal{L}_{\text{GWD}}$ and $\mathcal{L}_{\text{NLL}}$ are defined in (21) and (20), respectively. We set $\lambda > 0$ small so that the likelihood term stabilizes responsibilities.

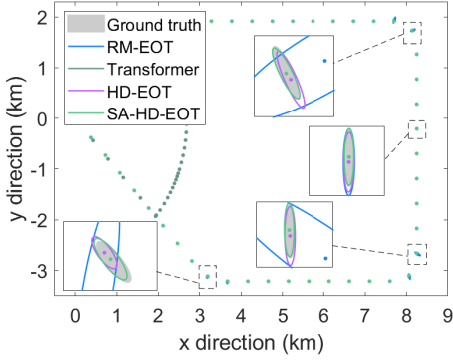


Fig. 2. The trajectories of the EO in a typical trial.

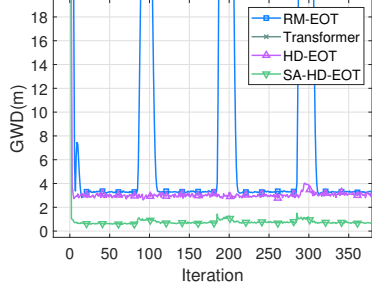


Fig. 3. The average GWDs of different approaches.

IV. EXPERIMENTAL RESULTS

We consider tracking a vehicle modeled as an ellipse with a major axis of 12 m and a minor axis of 3 m. The initial kinematic state $\mathbf{x}_0$ of the EO is $[0 \text{ m}, 0 \text{ m}, 13.9 \text{ m/s}, 13.9 \text{ m/s}]$, corresponding to an approximate constant velocity (CV). The EO then continues with CV motion and undergoes a 45° turn and two 90° turns at a constant turn rate (CT), with a fixed interval of $T = 5 \text{ s}$ [4], [5], [6].

At each time step k , a measurement set $\mathcal{Z}_k = \{\mathbf{z}_k^j\}_{j=1}^{n_k}$ is generated to emulate spatially non-uniform detections. Specifically, each scan is synthesized from $M = 4$ components consistent with (5). In the object-aligned frame, component centers are placed at $(\pm a/2, \pm b/2)$ and mapped to the global frame by heading rotation and translation to give $\{\mathbf{c}_{k,i}\}_{i=1}^4$ [8]. Component i is present with probability $P_{d,i}$ where $\mathbf{P}_d = [1.0, 0.85, 0.85, 1.0]$, and if present its cardinality follows $n_{k,i} \sim \text{Poisson}(\lambda_i)$ with $\boldsymbol{\lambda} = [80, 40, 40, 80]$. Conditioned on component i , measurements are sampled as $\mathbf{z} \sim \mathcal{N}(\mathbf{c}_{k,i}, \mathbf{X}_k/8 + \mathbf{R}_k)$. We consider that the measurement noise covariance and the kinematic process noise covariance are given by $\mathbf{R}_k = r^2 \mathbf{I}_2$ and $\mathbf{D}_k = \text{diag}(q_1^2, q_2^2)$, where $r = 0.8$, $q_1 = 0.01$, and $q_2 = 0.001$. The kinematic process noise follows (1) with $\mathbf{F}_k = \mathbf{I}_2$, and the extent evolution follows the Wishart dynamics in (2) with $\mathbf{A}_k = \mathbf{I}_2/\sqrt{20}$.

We compare the SA-HD-EOT against the RM-EOT [6], the HD-EOT [9], and a Transformer set-encoder baseline [9]. For training the learning-based methods, we construct the simu-

TABLE I
PARAMETER COUNT, PEAK GPU MEMORY, AND TOTAL RUNTIME.

Method	Params (k)	Mem (MB)	Total Time (s)
SA-HD-EOT (ours)	96.6	270.8	14.81
RM-EOT	—	202.8	25.18
HD-EOT	6.0	212.6	23.21
Transformer	1063	1082	17.09

lation dataset based on the above signal model by randomly generating the initial kinematic state and independently scaling the semi-major and semi-minor axes with random factors drawn from $\mathcal{U}(0.8, 1.2)$. For fair comparison, all methods share the same hardware/software setting, and all learning-based methods are trained on the same supervised tracking dataset using the staged strategy in Sec. III.

The proposed SA-HD-EOT is observed to outperform the RM-EOT, the HD-EOT, and the Transformer baseline in Figs. 2 and 3, achieving the lowest and most stable GWD across maneuvers. The RM-EOT exhibits pronounced error bursts around turn segments, which is consistent with biased scan-level single-cluster sufficient statistics under multimodal, non-uniform measurements. The Transformer baseline shows higher variance and occasional instability, indicating that purely data-driven regression can be less robust when the measurement set is strongly non-uniform. The HD-EOT improves over the RM-EOT by learning uncertain update terms, but it still relies on global centroid-style summaries and therefore retains residual bias under strong spatial non-uniformity. By contrast, the SA-HD-EOT leverages SAGNet to extract component-wise statistics that suppress between-component inflation, while mixture fusion stabilizes updates throughout maneuvers.

Additionally, Table I shows that SA-HD-EOT attains the lowest total runtime under the same implementation setting, with moderate memory usage. Compared with the HD-EOT, SA-HD-EOT introduces more parameters due to the slot-attention front-end, yet it remains much smaller than the Transformer baseline. Although RM-EOT has no learnable parameters, its online cost is dominated by scan-level statistic construction and RM matrix updates. By contrast, SA-HD-EOT replaces part of these analytical operations with a lightweight batched forward pass, which explains its lower total runtime in our implementation.

V. CONCLUSION

We propose the SA-HD-EOT, a slot-attention guided hybrid-driven extended object tracker for non-uniform measurement sets. The SAGNet decomposes variable-cardinality detections into component-wise statistics compatible with RM-style updates and fuses multiple HD-EOT sub-filters via mixture weights, enabling robust single-pass recursion. Experiments with maneuvering vehicles and non-uniform detections show that the proposed SA-HD-EOT achieves the lowest and most stable GWD among the compared methods, while attaining the lowest total runtime with moderate memory usage.

REFERENCES

- [1] K. Granström, M. Baum, and S. Reuter, "Extended object tracking: Introduction, overview and applications," *Journal of Advances in Information Fusion*, vol. 12, no. 2, pp. 139–174, 2017.
- [2] S. Haag, V. Duraisamy, J. Dickmann, and W. Koch, "Radar and lidar target signatures: An overview and a comparison of sensors and methods for the detection and tracking of extended objects," in *2018 21st International Conference on Information Fusion (FUSION)*, 2018, pp. 1746–1755.
- [3] L. Mihaylova, A. Y. Carmi, F. Septier, A. Gning, S. K. Pang, and S. J. Godsill, "Overview of bayesian sequential monte carlo methods for group and extended object tracking," *Digital Signal Processing*, vol. 25, pp. 1–16, 2014.
- [4] W. Koch, "Bayesian approach to extended object and cluster tracking using random matrices," *IEEE Transactions on Aerospace and Electronic Systems*, vol. 44, no. 3, pp. 1042–1059, 2008.
- [5] M. Feldmann, D. Fränken, and W. Koch, "Tracking of extended objects and group targets using random matrices," *IEEE Transactions on Signal Processing*, vol. 59, no. 4, pp. 1409–1420, 2011.
- [6] J. Lan and X. R. Li, "Tracking of extended object or target group using random matrix: New model and approach," *IEEE Transactions on Aerospace and Electronic Systems*, vol. 52, no. 6, pp. 2973–2989, 2016.
- [7] L. Zhang and J. Lan, "Extended object tracking using random matrix with skewness," *IEEE Transactions on Signal Processing*, vol. 68, pp. 5107–5121, 2020.
- [8] —, "Tracking of extended object using random matrix with non-uniformly distributed measurements," *IEEE Transactions on Signal Processing*, vol. 69, pp. 3812–3825, 2021.
- [9] R. Gan and W. Xia, "Extended object tracking: A DNN-aided approach," in *2025 Asia Pacific Signal and Information Processing Association Annual Summit and Conference (APSIPA ASC)*, 2025, pp. 1608–1614.
- [10] A. Vaswani, N. Shazeer, N. Parmar, J. Uszkoreit, L. Jones, A. N. Gomez, L. Kaiser, and I. Polosukhin, "Attention is all you need," in *Advances in Neural Information Processing Systems (NeurIPS)*, 2017, pp. 5998–6008.
- [11] M. Zaheer, S. Kottur, S. Ravanbakhsh, B. Póczos, R. Salakhutdinov, and A. Smola, "Deep sets," in *Advances in Neural Information Processing Systems (NeurIPS)*, 2017, pp. 3391–3401.
- [12] J. Lee, Y. Lee, J. Kim, A. R. Kosiorek, S. Choi, and Y. W. Teh, "Set transformer: A framework for attention-based permutation-invariant neural networks," in *International Conference on Machine Learning (ICML)*, 2019, pp. 3744–3753.
- [13] N. Shlezinger and Y. C. Eldar, "Model-based deep learning," *Foundations and Trends in Signal Processing*, vol. 17, no. 4, pp. 291–416, 2023.
- [14] G. Revach, N. Shlezinger, X. Ni, R. Klein, and Y. C. Eldar, "Kalmannet: Neural network aided kalman filtering for partially known dynamics," *IEEE Transactions on Signal Processing*, vol. 70, pp. 1532–1547, 2022.
- [15] X. Ni, G. Revach, and N. Shlezinger, "Adaptive kalmannet: Data-driven kalman filter with fast adaptation," in *IEEE International Conference on Acoustics, Speech and Signal Processing (ICASSP)*, 2024, pp. 5970–5974.
- [16] R. Klein, N. Mudrik, G. Revach, N. Shlezinger, and Y. C. Eldar, "Uncertainty in data-driven kalman filtering for partially known state space models," in *IEEE International Conference on Acoustics, Speech and Signal Processing (ICASSP)*, 2022, pp. 3194–3198.
- [17] Y. Fan, C. E. Ardic, M. Trinh-Hoang, and M. Pesavento, "Decentralized online direction-of-arrival estimation and tracking," in *2022 IEEE 12th Sensor Array and Multichannel Signal Processing Workshop (SAM)*, 2022, pp. 6–10.
- [18] B. Teng, X. Yuan, R. Wang, and S. Jin, "Bayesian user tracking for reconfigurable intelligent surface aided mmWave MIMO system," in *2022 IEEE 12th Sensor Array and Multichannel Signal Processing Workshop (SAM)*, 2022, pp. 201–205.
- [19] C. R. Qi, H. Su, K. Mo, and L. J. Guibas, "Pointnet: Deep learning on point sets for 3d classification and segmentation," in *IEEE Conference on Computer Vision and Pattern Recognition (CVPR)*, 2017, pp. 652–660.
- [20] F. Locatello, D. Weissenborn, T. Unterthiner, A. Mahendran, G. Heigold, J. Uszkoreit, A. Dosovitskiy, and T. Kipf, "Object-centric learning with slot attention," in *Advances in Neural Information Processing Systems (NeurIPS)*, 2020, pp. 11 525–11 538.
- [21] C. M. Bishop, *Pattern Recognition and Machine Learning*, ser. Information Science and Statistics. New York, NY: Springer, 2006.
- [22] A. P. Dempster, N. M. Laird, and D. B. Rubin, "Maximum likelihood from incomplete data via the EM algorithm," *Journal of the Royal Statistical Society: Series B*, vol. 39, no. 1, pp. 1–38, 1977.